

Oh time, don't fade away when I need you here

Oh please don't leave me now

@ Jari Mäenpää (2012), Wintersun – Time

```
#include <xc.h>
#include <libpic30.h>
#include <stdio.h>
#include <string.h>
#include "lcd.h"
#include "buttons.h"
#define TOTAL_TIME (300) // w sekundach

volatile unsigned int seconds = TOTAL_TIME;

int main(void) {
    char buffer[16]; // bufor na znaki do wyświetlenia na ekranie
    LCD_Initialize(); // inicjalizacja LCD
    unsigned int remainingSeconds = seconds;

    while(remainingSeconds > 1) {
        if (BUTTON_IsPressed ( BUTTON_S3 ) == true) {
            remainingSeconds = remainingSeconds + 30;
        } else if ((BUTTON_IsPressed ( BUTTON_S4 ) == true)) {
            remainingSeconds = remainingSeconds - 30;
        }

        unsigned char minutes = remainingSeconds / 60;
        unsigned char secs = remainingSeconds % 60;
        sprintf(buffer, "%02u:%02u", minutes, secs); // zapis czasu do bufora

        // wyświetlenie na ekranie
        LCD_ClearScreen();
        LCD_PutString(buffer, strlen(buffer));
        __delay32(4000000); //czekamy sekundę
        remainingSeconds--;
    }

    sprintf(buffer, "%12s", "koniec czasu");
    LCD_ClearScreen();
    LCD_PutString(buffer, strlen(buffer));
    __delay32(40000000); //czekamy 10 sekund
    return 0;
}
```